Dynamical Systems Homework #3

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Problem (Exercise 1). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. Recall the definition of the invertible extension $(\widetilde{X}, \widetilde{\mathcal{B}}, \widetilde{\mu}, \widetilde{T})$ from the exercise classes, and the factor map $\pi : \widetilde{X} \to X$. Show that $(\widetilde{X}, \widetilde{\mathcal{B}}, \widetilde{\mu}, \widetilde{T})$ has the following universal property: if $(Y, \mathcal{C}, \nu, S)$ is an invertible measure-preserving system and $\phi : Y \to X$ is a factor map, then there exists a unique factor map $\widetilde{\phi} : Y \to \widetilde{X}$ such that $\phi = \pi \circ \widetilde{\phi}$.*

Solution. Recall the definition of an invertible extension $(\widetilde{X}, \widetilde{\mathcal{B}}, \widetilde{\mu}, \widetilde{T})$:

- The space is $\widetilde{X} = \{x \in X^{\mathbb{Z}} : x_{n+1} = T(x_n)\ \forall_{n\in\mathbb{Z}}\}$.
- The transformation is $\widetilde{T}((x_k)_{k\in\mathbb{Z}}) = (x_{k+1})_{k\in\mathbb{Z}}$.
- For any $I \subseteq \mathbb{Z}$, $\{A_i\}_{i\in I} \subseteq \mathcal{B}$, the measure is:

$$\widetilde{\mu}\left(\left\{x \in \widetilde{X} : x_i \in A_i, i \in I\right\}\right) = \prod_{i\in I} \mu(A_i).$$

- $\widetilde{\mathcal{B}}$ is the smallest σ-algebra where $E \in \widetilde{\mathcal{B}}$ implies $T^{-1}(E) \in \widetilde{\mathcal{B}}$ and $\pi : (x_k)_{k\in\mathbb{Z}} \mapsto x_0$ is measurable.

Now, suppose we are given the factor map $\phi : Y \to X$. We need first to define the map $\widetilde{\phi} : Y \to \widetilde{X}$. The map is quite simple:

$$\widetilde{\phi} : Y \to \widetilde{X}, \quad y \mapsto (\phi \circ S^n(y))_{n\in\mathbb{Z}}.$$

Let us check whether the map is well-defined. First, is $\widetilde{\phi}(y)$ an element of $\widetilde{X}$ in the first place? We check whether $T(x_n) = x_{n+1}$:

$$T(x_n) = T \circ \phi \circ S^n(y) = \phi \circ S \circ S^n(y) = \phi \circ S^{n+1}(y) = T(x_{n+1}).$$

To show that $\widetilde{\phi}$ is a factor map, we check *(a)* compatibility with π, *(b)* commutation with dynamics, and *(c)* measure preservation.

Part (a). We need $\pi \circ \widetilde{\phi} = \phi$. Expanding definition of π, we have $\pi \circ \widetilde{\phi}(y) = \phi \circ S^0(y) = \phi(y)$.

Part (b). We need $\widetilde{\phi} \circ S = \widetilde{T} \circ \widetilde{\phi}$. On the one hand, $(\widetilde{\phi} \circ S(y))_k = \phi \circ S^k \circ S(y) = \phi \circ S^{k+1}(y)$. On the other hand, $(\widetilde{T} \circ \widetilde{\phi}(y))_k = \widetilde{T}((\phi \circ S^n(y))_{n\in\mathbb{Z}})_k = \phi \circ S^{k+1}(y)$.

Part (c). We must show $\widetilde{\phi}_*\nu = \widetilde{\mu}$. It suffices to check this on the basic cylinder sets, i.e. $C_k(A) := \{x \in \widetilde{X} : x_k \in A\}$ for $A \in \mathcal{B}$. By the definition, $\widetilde{\mu}(C_k) = \mu(A)$. Now, compute the preimage under $\widetilde{\phi}$:

$$\widetilde{\phi}^{-1}(C_k(A)) = \{y \in Y : \widetilde{\phi}(y)_k \in A\} = \{y \in Y : \phi \circ S^k(y) \in A\} = S^{-k} \circ \phi^{-1}(A).$$

Now measure this set: $\nu(\widetilde{\phi}^{-1}(C_k(A))) = \nu(S^{-k} \circ \phi^{-1}(A))$. Since S is measure-preserving, this simplifies to $\nu(\widetilde{\phi}^{-1}(C_k(A))) = \nu(\phi^{-1}(A)) = \mu(A) = \widetilde{\mu}(C_k(A))$.

Finally, we show uniqueness. Suppose we have constructed another map $\psi : Y \to \widetilde{X}$ satisfying $\pi \circ \psi = \phi$. From the projection condition, $\psi(y)_0 = \phi(y)$. Since ψ is a factor map, $\psi \circ S^k(y) = \widetilde{T}^k \circ \psi$. Look at zero coordinates at both sides (*i.e.*, apply π): $\phi \circ S^k(y) = \psi(y)_k$. Thus, this forces ψ to be the map we defined earlier.

Problem (Exercise 2). *For each $\alpha, \beta \in \mathbb{R}$, determine whether or not the system $(\mathbb{R}/\mathbb{Z}, (x,y) \mapsto (x+\alpha, y+\beta))$ is (a) ergodic, (b) weakly mixing.*

Solution. We use Fourier techniques. First, we notice that this system is never weakly mixing. Indeed, recall that the system is weakly mixing if and only if the Koopman operator $U_T : f \mapsto f \circ T$ has no non-trivial eigenfunctions. However, take $\chi_{n,m}(x,y) := e^{2\pi i(nx+my)}$ (orthonormal basis of $L^2(\mathbb{R}^2/\mathbb{Z}^2)$). Then,

$$U_T \chi_{n,m} = e^{2\pi i(n\alpha+m\beta)} \chi_{n,m},$$

so $\{\chi_{n,m}\}_{n,m\in\mathbb{Z}}$ are non-trivial eigenfunctions of U_T.

For ergodicity, let $f(x,y) = \sum_{n,m\in\mathbb{Z}} \widehat{f}_{n,m}\chi_{n,m}(x,y)$. If the system is ergodic, $U_T f = f$ has only constant solutions (i.e. $\widehat{f}_{n,m} = 0$ for $(n,m) \neq (0,0)$). So,

$$\sum_{n,m\in\mathbb{Z}} \widehat{f}_{n,m}\chi_{n,m}(\alpha,\beta)\chi_{n,m}(x,y) = \sum_{n,m\in\mathbb{Z}} \widehat{f}_{n,m}\chi_{n,m}(x,y),$$

which forces $\widehat{f}_{n,m}\chi_{n,m}(\alpha,\beta) = \widehat{f}_{n,m}$. Thus we see that the given system is ergodic iff $e^{2\pi i(n\alpha+m\beta)} = 1$ only when $(n,m) = (0,0)$. This happens precisely when $\{1, \alpha, \beta\}$ are rationally independent.

Problem (Exercise 3). *For each $a, b, c, d \in \mathbb{Z}$ with $ad \neq bc$, determine whether or not the system $(\mathbb{R}^2/\mathbb{Z}^2, (x,y) \mapsto (ax+by, cx+dy))$ is (a) ergodic, (b) weakly mixing.*

Solution. We view our map as $T : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \mathbf{A}\begin{pmatrix} x \\ y \end{pmatrix}$ where $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Here, condition $ad \neq bc$ means that $\det \mathbf{A} \neq 0$. Note that our system is a compact abelian group, thus we might apply *Proposition 3.26* from the lecture notes: it is ergodic iff for all $N \in \mathbb{N}$ and non-unit characters $\chi_{n,m}$, we have $U_T^N \chi_{n,m} \neq \chi_{n,m}$. Here, non-unit characters are precisely $\chi_{n,m}(x,y) = e^{2\pi i(nx+my)}$ as in the previous exercise (since characters are precisely an orthonormal basis for $L^2(\mathbb{R}^2/\mathbb{Z}^2)$ thanks to *Proposition 3.20.*) Now notice:

$$U_T \chi_{n,m} = e^{2\pi i(n(ax+by)+m(cx+dy))} = e^{2\pi i((an+cm)x+(bn+dm)y)} = \chi_{(n,m)\mathbf{A}},$$

Consequently, $U_T^N \chi_{n,m} = \chi_{(n,m)\mathbf{A}^N} = \chi_{n,m}$ only when $\mathbf{A}^N = \mathbb{I}_2$. From the linear algebra, we now that this never happens for every $N \in \mathbb{N}$ iff neither eigenvalue of $\mathbf{A}$ is a root of unity.

Now we consider the weak mixing. The system is weakly mixing iff U_T has no non-trivial eigenfunctions. Thus, substitute everything into $U_T f = \lambda f$ assuming $f(x,y) = \sum_{n,m\in\mathbb{Z}} \widehat{f}_{n,m}\chi_{n,m}(x,y)$:

$$\sum_{n,m\in\mathbb{Z}} \widehat{f}_{n,m}\chi_{(n,m)\mathbf{A}}(x,y) = \sum_{n,m\in\mathbb{Z}} \lambda\widehat{f}_{n,m}\chi_{n,m}(x,y),$$

which forces $\widehat{f}_{(n,m)\mathbf{A}} = \lambda\widehat{f}_{n,m}$. Since $|\lambda| = 1$, this forces all coefficients with indices in orbit $\{(n,m)\mathbf{A}^N\}_{N\in\mathbb{Z}}$ to be equal. However, notice that if this orbit is infinite, this

would imply that all but $\widehat{f}_{0,0}$ coefficients are zero as $\sum_{n,m\in\mathbb{Z}} |\widehat{f}_{n,m}|^2 < \infty$, and thus the system is weakly mixing. The orbit is *finite* when $(n,m)\mathbf{A}^N = (n,m)$ for some $N \in \mathbb{N}$. But notice that this is exactly the condition when ergodicity fails. Thus, the system is weakly mixing iff no eigenvalue of $\mathbf{A}$ is a root of unity.

Problem (Exercise 4). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and let $1 \leq p < \infty$. Show that for each $f \in L^p(X,\mu)$ we have*

$$\frac{1}{N}\sum_{n=0}^{N-1} f \circ T^n \xrightarrow[N\to\infty]{L^p} \widetilde{f},$$

for some invariant $\widetilde{f} \in L^p(X,\mu)$.

Solution. In what follows, as suggested, I mimic the idea of Corollary 2.22 from the textbook. For convenience, further denote ergodic averages by

$$A_N g := \frac{1}{N}\sum_{n=0}^{N-1} g \circ T^n.$$

By the L^2 Mean Ergodic Theorem, we know that for any $g \in L^\infty(X,\mu) \subseteq L^2(X,\mu)$, the ergodic averages $A_N g$ converge in $L^2(X,\mu)$ to some T-invariant $\widetilde{g} \in L^2(X,\mu)$. We first claim that $\widetilde{g} \in L^\infty(X,\mu)$. Indeed, $\|A_N g\|_\infty \leq \|g\|_\infty$ and thus

$$|\langle A_N g, \chi_B\rangle| \leq \|g\|_\infty \mu(B) \text{ for all } B \in \mathcal{B}.$$

Since $A_N g \to \widetilde{g}$ in $L^2(X,\mu)$, we obtain $|\langle \widetilde{g}, \chi_B\rangle| \leq \|g\|_\infty \mu(B)$ for $B \in \mathcal{B}$, implying $\|\widetilde{g}\|_\infty \leq \|g\|_\infty$, as required.

Now we establish $A_N g \to \widetilde{g}$ in $L^p(X,\mu)$. Consider two cases.

Case #1. $1 \leq p \leq 2$. In such case $\|\cdot\|_p \leq \|\cdot\|_2$, and thus L^2 convergence immediately implies L^p convergence.

Case #2. $p > 2$. Here, we use the fact that both $A_N g$ and $\widetilde{g}$ are bounded by $\|g\|_\infty$:

$$\int_X |A_N g - \widetilde{g}|^p d\mu = \int_X |A_N g - \widetilde{g}|^{p-2} |A_N g - \widetilde{g}|^2 d\mu \leq (2\|g\|_\infty)^{p-2} \|A_N g - \widetilde{g}\|_2^2.$$

Since $\|A_N g - \widetilde{g}\|_2 \to 0$, we deduce $A_N g \to \widetilde{g}$ in $L^p(X,\mu)$.

Thus, we have demonstrated that the theorem holds for a dense set of functions $L^\infty(X,\mu) \subseteq L^p(X,\mu)$. Now we extend the argument to the whole $L^p(X,\mu)$. Let $f \in L^p(X,\mu)$ and fix $\varepsilon > 0$. Since $L^\infty(X,\mu)$ is dense in $L^p(X,\mu)$, choose $g \in L^\infty(X,\mu)$ such that $\|f - g\|_p < \varepsilon$. Now note that $\|f \circ T\|_p = \|f\|_p$. Consequently, $\|A_N f\|_p \leq \|f\|_p$. This gives $\|A_N f - A_N g\|_p \leq \|f - g\|_p < \varepsilon$. By the previous result, $A_N g$ converges in $L^p(X,\mu)$, meaning there exists N_0 such that $\|A_N g - A_M g\|_p < \varepsilon$ for all $N, M \geq N_0$. Thus,

$$\|A_N f - A_M f\|_p \leq \|A_N f - A_N g\|_p + \|A_N g - A_M g\|_p + \|A_M g - A_M f\|_p < 3\varepsilon$$

whenever $N, M \geq N_0$. Thus, $A_N f$ form a Cauchy sequence in $L^p(X,\mu)$, meaning that it converges to some $\widetilde{f} \in L^p(X,\mu)$.

What is left to show that $\widetilde{f}$ is T-invariant. Since $U_T(A_N f) = A_N f \circ T$, we have

$$\|A_N f \circ T - A_N f\|_p = \left\| \frac{1}{N} f \circ T^N - \frac{1}{N} f \right\|_p \leq \frac{2}{N} \|f\|_p \xrightarrow[N\to\infty]{} 0.$$

Since $A_N f \to \widetilde{f}$ in $L^p(X, \mu)$ and U_T is continuous, the left side converges to $\|\widehat{f} \circ T - \widehat{f}\|_p$ which was shown to be 0. Thus, $\widetilde{f} \circ T = \widetilde{f}$.

Problem (Exercise 5). *For $k \geq 1$, let $\mathcal{Z}_k = (\mathbb{Z}/k\mathbb{Z}, \mathcal{P}(\mathbb{Z}/k\mathbb{Z}), \lambda_k, x \mapsto x+1)$ denote the cyclic shift on k elements. Let $\mathcal{X} = (X, \mathcal{B}, \mu, T)$ be a measure-preserving system. Define the product $\mathcal{X} \times_* \mathcal{Z}_k$ as the dynamical system under natural probability space and transformation $T^{(k)}$ given by*

$$T^{(k)}(x, i) := \begin{cases} (x, i+1), \ 0 \leq i < k-1, \\ (T(x), 0), \ i = k-1. \end{cases}$$

(a) Show that $\mathcal{X} \times_* \mathcal{Z}_k$ is ergodic if and only if $\mathcal{X}$ is ergodic.
(b) Show that $\mathcal{X} \times_* \mathcal{Z}_k$ is totally ergodic if and only if $\mathcal{X}$ is totally ergodic and $k = 1$.
(c) Show that each measure-preserving system $\mathcal{Y} = (Y, \mathcal{C}, \nu, S)$ that is ergodic but not totally ergodic is isomorphic to some system of the form $\mathcal{X} \times_* \mathcal{Z}_k$ with $k > 1$.

Solution.

Part (a). We will use Fourier-based approach, as usual. Recall that the orthonormal basis for $L^2(\mathbb{Z}/k\mathbb{Z})$ is $\{\chi_m\}_{m<k}$ where $\chi_m(j) = e^{2\pi i m j/k}$. Therefore, any function $F \in L^2(X \times (\mathbb{Z}/k\mathbb{Z}))$ can be uniquely expanded as follows:

$$F(x, j) = \sum_{k=0}^{m-1} f_m(x)\chi_m(j) \text{ where } f_0, \ldots, f_{m-1} \in L^2(X).$$

Now consider the equation $U_{T^{(k)}} F = F$. For all $j < k-1$, this reduces to the equation $F(x, j+1) = F(x, j)$. Since in such case $F(x, j) = F(x, 0)$ for all $j < k-1$, F does not depend on the second coordinate. Thus, let $F(x, \cdot) \equiv f(x) \in L^2(X, \mu)$. The equation for $j = k-1$ then corresponds to $f(Tx) = f(x)$ or $U_T f = f$.

From here, the result is clear: if $\mathcal{X}$ is ergodic, f must be a constant, forcing F to be a constant, so $\mathcal{X} \times_* \mathcal{Z}_k$ is ergodic. Conversely, any invariant f generates the invariant F, which must be constant, making f constant as well, thus $\mathcal{X}$ is ergodic.

Part (b). Recall that total ergodicity is equivalent to stating that Koopman operator has no eigenvalues other than 1. Suppose $k > 1$. Let $\lambda := e^{2\pi i/k}$. We claim that it is an eigenvalue of $U_{T^{(k)}}$ with the corresponding eigenfunction $g(x, j) = \lambda^j$. Indeed, for $j < k-1$, $U_{T^{(k)}} g(x, j) = g(x, j+1) = \lambda^{j+1} = \lambda g(x, j)$. For $j = k-1$, we have $U_{T^{(k)}} g(x, k-1) = g(T(x), 0) = 1$. Notice that this is precisely $\lambda g(x, k-1)$ since $\lambda \cdot \lambda^{k-1} = \lambda^k = 1$ (as λ is a primitive kth root of unity).

Thus, total ergodicity is possible only for $k = 1$, for which the statement is trivial.

Part (c). Since $\mathcal{Y}$ is ergodic but not totally ergodic, the Koopman operator U_S on $L^2(Y, \nu)$ must have a non-trivial eigenvalue which is a root of unity. Let $k > 1$ be

the smallest positive integer such that $\lambda = e^{2\pi i/k}$ is an eigenvalue of U_S. Then, there exists an eigenfunction $h \in L^2(Y, \nu)$ such that $h \circ S = e^{2\pi i/k} h$.

This equation results in an important observation.

Proposition 0.1. *For any measure-preserving system $\mathcal{Y} = (Y, \mathcal{C}, \nu, S)$ that is ergodic but not totally ergodic, the following statements are true:*

(a) There exists a non-trivial $\varphi \in L^2(Y, \nu)$ such that $\varphi^k = 1$ ν-a.e. for some $k > 1$.

(b) Using notation from (a), Y can be partitioned into k sets $Y_0, \ldots, Y_{k-1}$ where $Y_j := \{y \in Y : \varphi(y) = e^{2\pi i j/k}\}$. Moreover, $S(Y_j) = Y_{(j+1) \bmod k}$.

Statement *(a)* follows from the fact that as $\mathcal{Y}$ is ergodic and $|h \circ S| = |h|$, $|h|$ is a constant. Without loss of generality, assume this constant is 1 for a function φ. Moreover, since $\varphi^k \circ S = (e^{2\pi i/k})^k \varphi^k = \varphi^k$, the function φ^k is a constant. Again, since $|\varphi| = 1$, without loss of generality we might assume $\varphi^k = 1$ ν-a.e.

Given *(a)*, the statement *(b)* is now trivial.

This proposition allows us to define the following measure-preserving system: $\mathcal{X} := (Y_0, k\nu|_{Y_0}, S^k|_{Y_0})$. The isomorphism is then defined as

$$\phi : \mathcal{X} \times (\mathbb{Z}/k\mathbb{Z}) \to Y, \quad (y_0, j) \mapsto S^j(y_0).$$

First, this is a measure-preserving map: if one takes an open $E \in \mathcal{C}$ (without loss of generality assume $E \subseteq Y_j$ for some j), then $\phi^{-1}(E) = S^{-j}(E) \times \{j\}$. Thus, $k\nu|_{Y_0}(\phi^{-1}(E)) = \nu(S^{-j}(E)) = \nu(E)$. It is also a bijection since the inverse map ϕ^{-1} is given by mapping $y \in Y$ to (S^{-j}, j) where $\varphi(y) \in Y_j$. Finally, $\phi \circ T^{(k)} = S \circ \phi$ also checks fine.

[Vri14] J. Vries. *Topological Dynamical Systems: An Introduction to the Dynamics of Continuous Mappings*. De Gruyter Studies in Mathematics. De Gruyter, 2014. URL: https://books.google.com.ua/books?id=hyroBQAAQBAJ.